

Document Management API Chernyx AI Labs — Design

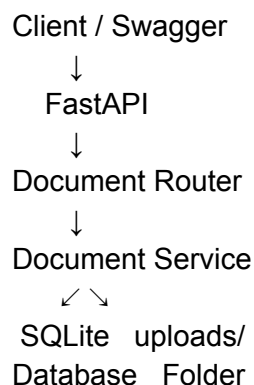
1. Overview

A REST API built using **Python** and **FastAPI** to manage PDF documents.

The API supports:

- PDF upload
- File size validation (maximum 5 MB)
- SHA-256 duplicate detection
- Local file storage
- SQLite metadata storage
- Document listing, retrieval, and deletion

2. Architecture



- **FastAPI:** Handles HTTP requests and responses.
- **Document Router:** Handles API endpoints and routes requests.
- **Document Service:** Contains validation and document-management business logic.
- **SQLite:** Stores document metadata.
- **uploads:** Stores the actual PDF files.

3. Database Design

The `documents` table stores:

Field	Description
id	Unique document ID

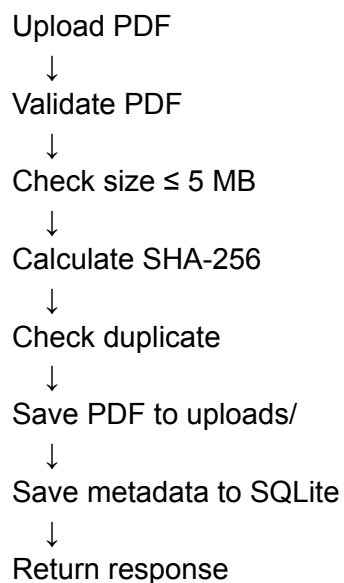
filename	Original PDF filename
file_size	File size in bytes
file_hash	SHA-256 hash
status	Document status
created_at	Upload timestamp
updated_at	Last update timestamp

`file_hash` is unique to detect duplicate uploads.

4. API Endpoints

Method	Endpoint	Purpose
POST	<code>/documents</code>	Upload a PDF
GET	<code>/documents</code>	List all documents
GET	<code>/documents/{id}</code>	Get document metadata
DELETE	<code>/documents/{id}</code>	Delete a document

5. Upload Flow



6. Validation & Error Handling

- Invalid/non-PDF file → **400 Bad Request**
- File larger than 5 MB → **413 Payload Too Large**
- Duplicate file → **409 Conflict**
- Document not found → **404 Not Found**
- Successful upload → **201 Created**
- Successful deletion → **204 No Content**

7. Testing

Pytest will cover:

- Valid PDF upload
- Invalid file
- File larger than 5 MB
- Duplicate upload
- List documents
- Get document
- Delete document
- Missing document

8. Documentation

FastAPI automatically provides Swagger/OpenAPI documentation at:

[/docs](#)

9. Technology Stack

- Python
- FastAPI
- SQLite
- Pytest
- Uvicorn

Actual PDF files are stored locally in [uploads/](#), while only their metadata is stored in SQLite.